

Section : Common

Q.1 Let $A = [a_{ij}]_{3 \times 2}$ and $B = [b_{ij}]_{3 \times 4}$ be two matrices.

Then the order of the matrix $(A^T \cdot B)^T$ is:

1. 2×4
2. 4×2
3. 3×4
4. 4×3

Options 1. 1

2. 2
3. 3
4. 4

Q.2 If $y = ax^2 + bx$ has minima at $x = 2$ and the minimum value is -12 , then which of the following are correct?

- (A) $a = 3$
- (B) $a = -3$
- (C) $b = 12$
- (D) $b = -12$

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (A) and (D) only
3. (A) and (C) only
4. (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.3 If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $A^2 = KA - 2I$, then the value of K is

1. 1
2. 2
3. -2
4. 0

Options 1. 1

2. 2
3. 3
4. 4

Q.4

If $y = t - \frac{1}{t}$ and $x = t + \frac{1}{t}$, then $\frac{dy}{dx}$ is equal to

1. $\frac{t^2 - 1}{t^2 + 1}$
2. $\frac{t^2 + 1}{t^2 - 1}$
3. $\frac{t - 1}{t + 1}$
4. $\frac{t + 1}{t - 1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The real valued function $f(x) = x^{15} + 5x^9 + 10$ is increasing for ____.

1. all positive real values of x
2. all negative real values of x
3. all non-positive real values of x
4. all real values of x

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Match List-I with List-II

List-I

Differential Equation

(A) $\frac{dy}{dx} = \frac{y}{x}, x \neq 0$

(B) $x dx - y dy = 0; y \neq 0, x \neq 0$ (II) $x^2 - y^2 = c, c$ is an arbitrary constant

(C) $\frac{(x^2 - 1)}{y^2 + 1} \frac{dy}{dx} = 1$

(D) $2dx + 3dy = 0$

List-II

General solution

(I) $y = cx, c$ is an arbitrary constant

(III) $2x + 3y = c, c$ is an arbitrary constant

(IV) $(x^3 - y^3) = c + 3(x + y), c$ is an arbitrary constant

Choose the correct answer from the options given below:

- (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
- (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 For $x > e$,

$\int \frac{dx}{x - \sqrt{x}}$ is equal to

- $-2 \log_e |\sqrt{x} - 1| + C : C$ is an arbitrary constant
- $4 \log_e |\sqrt{x} - 1| + C : C$ is an arbitrary constant
- $\log_e |\sqrt{x} - 1| + C : C$ is an arbitrary constant
- $2 \log_e |\sqrt{x} - 1| + C : C$ is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The area (in square units) of the region enclosed between the lines $x + y = 2, x = 0, x = 3$ and x - axis is equal to

- $\frac{6}{7}$
- $\frac{7}{12}$
- $\frac{5}{2}$
- 7

Options 1. 1

2. 2
3. 3
4. 4

Q.9 If the random variable X has the following probability distribution:

X 0 1 2 Otherwise

$P(X)$ K 3K 5K 0

then K is equal to

- $\frac{4}{9}$
- $\frac{2}{9}$
- $\frac{1}{9}$
- $\frac{8}{9}$

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Match List-I with List-II

List-I

List-II

(Definite integral) (Value)

(A) $\int_1^e \frac{\log x}{x} dx$ (I) 4

(B) $\int_{-2}^2 x^3(1-x^2) dx$ (II) $\frac{1}{2}$

(C) $\int_1^2 x dx$ (III) 0

(D) $\int_{-2}^2 |x| dx$ (IV) $\frac{3}{2}$

Choose the correct answer from the options given below:

1. (A) – (II), (B) – (III), (C) – (IV), (D) – (I)
2. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)
3. (A) – (I), (B) – (III), (C) – (IV), (D) – (II)
4. (A) – (III), (B) – (II), (C) – (I), (D) – (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.11 The corner points of the bounded feasible region determined by the system of linear constraints are (0, 0), (5, 0), (6, 5), (6, 8), (4, 10), (0, 8). Let $z = 3x - 4y$ be the objective function. Then the minimum value of z occurs at:

1. (0, 0)
2. (0, 8)
3. (5, 0)
4. (4, 10)

Options 1. 1

2. 2
3. 3
4. 4

Q.12 If the objective function $z = px + qy$ has its maximum value at the points (2, 1) and (0, 6), then the relationship between p and q is:

1. $5p = 2q$
2. $2p = 5q$
3. $p = 2q$
4. $2p = q$

Options 1. 1

2. 2
3. 3
4. 4

Q.13 The number of arbitrary constants in the general solution of a differential equation with degree 1 and order 3, is

1. 1
2. 2
3. 3
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.14

The inverse of the matrix $A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$ is

1. $\frac{1}{64} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}$

2. $\frac{1}{64} \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$

3. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}$

4. $\frac{1}{64} \begin{bmatrix} \frac{1}{32} & 0 & 0 \\ 0 & \frac{1}{16} & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.15 Let M_{ij} and A_{ij} denote respectively minors and co-factors of the element in the i th row and j th column of the matrix

$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 3 \\ 4 & -1 & 0 \end{bmatrix}$. Then

(A) $M_{32} = 6$

(B) $M_{23} = 9$

(C) $A_{32} = -6$

(D) $A_{23} = -9$

Choose the correct answer from the options given below:

1. (A) and (C) only

2. (A), (C) and (D) only

3. (B) and (D) only

4. (A) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Section : Applied Mathematics

Q.16 Consider the following hypothesis test :

$$H_0 : \mu = 15$$

$$H_a : \mu \neq 15$$

A sample of 50 provided a sample mean of 14.15. If the sample standard deviation is 3, then the value of the test statistic in t-test is

1. -1

2. -2

3. 1

4. 2

Options 1. 1

2. 2

3. 3

4. 4

Q.17 The demand function for a commodity is $p = 35 - 2x - x^2$, then the consumer's surplus at equilibrium price $p_0 = 20$ is

1. 20

2. 24

3. 27

4. 28

Options 1. 1

2. 2

3. 3

4. 4

Q.18 The solution set of the inequation $|2x - 3| \leq 4$ is

1. $[\frac{1}{2}, \frac{7}{2}]$

2. $[-\frac{1}{2}, \frac{7}{2}]$

3. $[-\frac{1}{2}, \frac{5}{2}]$

4. $[-\frac{7}{2}, \frac{7}{2}]$

Options 1. 1

2. 2

3. 3

4. 4

Q.19 If the matrix $\begin{bmatrix} 1 & 3 & 9 \\ 1 & x & x^2 \\ 4 & 6 & 9 \end{bmatrix}$ is a singular matrix, where $x \in \mathbb{N}$ (the set of natural numbers), then x is equal to

1. 2

2. 4

3. 3

4. 5

Options 1. 1

2. 2

3. 3

4. 4

Q.20 \[\text{If for the following data:} \]

YEAR(X)	2000	2001	2002	2003	2004	2005	2006
PRODUCTION (IN TONNES)(Y)	40	45	46	42	47	50	46

the equation of the straight line trend is $y = 45.143 + 1.036(x - 2003)$, then the trend value for the year 2004 is

1. 46.18
2. 45.18
3. 48.18
4. 50.18

Options 1. 1

2. 2
3. 3
4. 4

Q.21 The number of years required for a sum of money to get tripple at the effective rate of 4\% is : (Given $(1.04)^{28} = 3$)

1. 7 years
2. 14 years
3. 28 years
4. 56 years

Options 1. 1

2. 2
3. 3
4. 4

Q.22 If a random variable χ follows Poisson's distribution such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, then sum of the mean and variance of χ is

1. 4
2. 6
3. 1
4. 2

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Which of the following are correct about the linear programming problem(LPP)?

- (A) The objective function is always linear.
- (B) A corner point of a feasible region is a point in the region which is the intersection of two boundary lines.
- (C) The common region determined by all the linear constraints of a LPP is called the feasible region.
- (D) If an LPP admits optimal solution at two points then its optimal values occurs at an infinite number of points.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (C) and (D) only
3. (A), (B), (C) and (D)
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The mean of the number of heads in a simultaneous toss of three coins is

1. $\frac{3}{2}$
2. $\frac{1}{2}$
3. 2
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Which of the following is correct about the compound annual growth rate (CAGR)?

1. CAGR is calculated by only the final value of an investment.
2. CAGR is calculated by the final and beginning value of an investment.
3. CAGR describes the effect of simple interest.
4. CAGR is calculated by taking arithmetic mean of series of returns.

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Ram can row a boat in still water at the rate of 6 km/hr. He finds that it takes him twice as long to row upstream of a river as to row downstream of the same river, then the speed of the current of the river is

1. 5 km/hr
2. 4 km/hr
3. 3 km/hr
4. 2 km/hr

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The point on the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ at which the tangent to the curve is parallel to the x-axis is

1. (0,3)
2. (2,0)
3. (-2,0)
4. (2,3)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 In a linear programming problem (LPP), the maximum value of the objective function $Z = 2x + 5y$ subjected to the constraints:

$$2x + 3y \leq 6$$

$$2x + y \leq 4$$

$$x, y \geq 0$$

is

1. 12
2. 10
3. 8
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.29 A die is thrown 6 times. If getting an odd number is a success, then the probability of at least 5 successes is

1. $\frac{63}{64}$
2. $\frac{9}{64}$

3. $\frac{7}{64}$
4. $\frac{1}{64}$

Options 1. 1
2. 2
3. 3
4. 4

Q.30 If $\begin{bmatrix} x-y & t \\ 2x-y & w \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}$, then $2x + y + 3t + w$ is equal to

1. 9
2. 11
3. 12
4. 10

Options 1. 1
2. 2
3. 3
4. 4

Q.31 The least positive integer x satisfying $29 \equiv x \pmod{5}$ is

1. 1
2. 3
3. 4
4. 5

Options 1. 1
2. 2
3. 3
4. 4

Q.32 Which of the following statements are correct?

- (A) In the sinking fund a fixed amount at regular intervals is deposited.
(B) Sinking fund is a long-term account which can be closed any time.
(C) In a saving account, any amount, any time can be deposited.
(D) Sinking fund can be used only for the purpose it was created.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (B), (C) and (D) only
4. (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

Q.33 If we reject the null hypothesis when it is true, we might be making

1. Type-I error
2. Type-II error
3. Sampling error
4. Calculation error

Options 1. 1
2. 2
3. 3
4. 4

Q.34 If water is mixed with milk to gain 20% by selling the mixture at cost price. Then ratio of water to milk in mixture, is :

1. 4:5
2. 2:5
3. 1:5
4. 5:2

Options 1. 1

2. 2
3. 3
4. 4

Q.35 If $f(x) = x^2 - 4x + 13, x \in \mathbb{R}$, then which of the following are correct?

- (A) $x = 2$ is a stationary point of $f(x)$.
- (B) $f(x)$ is increasing function on $(2, \infty)$.
- (C) $f(x)$ have maxima at $x = 2$.
- (D) $f(2) = 9$.

Choose the **correct** answer from the options given below:

1. (A), (C) and (D) only
2. (B), (C) and (D) only
3. (A), (B) and (C) only
4. (A), (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.36 A mobile phone costing ₹50000 has a useful life of 5 years. If the annual depreciation is ₹5000, then by using a linear method, its scrap value is

1. ₹25000
2. ₹30000
3. ₹20000
4. ₹15000

Options 1. 1

2. 2
3. 3
4. 4

Q.37 The rise and fall of share market is an example of

1. Seasonal trend
2. Cyclical trend
3. Secular trend
4. Irregular trend

Options 1. 1

2. 2
3. 3
4. 4

Q.38 There are 50 telephone lines in an exchange. The probability that any one of them will be busy is 0.1, then the probability that all the lines are busy?

1. $\frac{e^{-5} \times 5^{50}}{5!}$
2. $\frac{e^{-5} \times 5^{50}}{50!}$
3. $\frac{e^{-50} \times 50^{50}}{50!}$
4. $\frac{e^{-50} \times 50^5}{50!}$

Options 1. 1

2. 2

3. 3

4. 4

Q.39 A pipe A can fill a tank in 4 hours. There are two outlet pipes, B and C, from the tank which can empty it in 8 and 10 hours respectively. If all the three pipes are opened simultaneously, how long will it take to fill the tank?

1. 30 hours
2. 40 hours
3. 45 hours
4. 50 hours

Options 1. 1

2. 2
3. 3
4. 4

Q.40 The system of equations

$$x + ky = 0$$

$$3x + 5y = 0$$

has infinitely many solutions, then k is equal to

1. $\frac{3}{5}$
2. $\frac{5}{3}$
3. $-\frac{5}{3}$
4. $-\frac{3}{5}$

Options 1. 1

2. 2
3. 3
4. 4

Q.41 For the given five values 18, 25, 20, 36, 49, the three years moving averages are

1. 21, 27, 36
2. 21, 27, 35
3. 21, 28, 35
4. 22, 28, 34

Options 1. 1

2. 2
3. 3
4. 4

Q.42 The differential equation representing the curve $y = e^{2x}(a + bx)$, where a, b are arbitrary constants is

1. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$
2. $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} - 4y = 0$
3. $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$
4. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 4y = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Which of the following are correct?

- (A) The probability curve in t-distribution is symmetric about the line $t=0$.
- (B) t-axis is an asymptote of the curve.

(C) The variable t of t -distribution ranges from $-\infty$ to ∞ .

(D) As the number of degrees of freedom increases, the t -distribution curve moves closer to the binomial distribution.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.44

If A is a square matrix of order 3 such that $A(\text{adj } A) = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$, then $|A|$ is equal to

1. 2
2. -8
3. 8
4. -2

Options 1. 1
2. 2
3. 3
4. 4

Q.45 If A and B are symmetric matrices of same order, then which of the following are correct?

- (A) $AB-BA$ is a skew-symmetric matrix.
- (B) $AB+BA$ is a skew-symmetric matrix.
- (C) $AB^T - BA^T$ is a skew-symmetric matrix.
- (D) $AB+BA$ is a symmetric matrix.

Choose the \textbf{correct} answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (C) and (D) only
4. (A) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.46 In a game of 100 points, A can give B 20 points and C 28 points. How many points can B give C?

1. 25 points
2. 20 points
3. 15 points
4. 10 points

Options 1. 1
2. 2
3. 3
4. 4

Q.47 The demand for a certain product is represented by the function $p = 300 + 25x - x^2$ (in rupees), where x is the number of units demanded and p is the price per unit, then the marginal revenue when 15 units are sold, is

1. ₹675

2. ₹375
3. ₹1050
4. ₹775

Options 1. 1
2. 2
3. 3
4. 4

Q.48 Anshu takes a personal loan of ₹10,00,000 at the rate of 12\% per annum for 2 years, then the EMI by using flat rate method is

1. ₹51,666.67
2. ₹52,666.67
3. ₹50,666.67
4. ₹53,666.67

Options 1. 1
2. 2
3. 3
4. 4

Q.49 If $(t_1, y_1), (t_2, y_2), (t_3, y_3), \dots, (t_n, y_n)$ denote the time series and y_t are the trend values of the variable y , then

$\sum (y - y_t)$, the sum of deviations of y from their corresponding trend value is equal to:

1. 1
2. -1
3. 0
4. 2

Options 1. 1
2. 2
3. 3
4. 4

Q.50 What sum of money is needed to invest now, so as to get ₹5000 at the beginning of every month forever, if the money is worth 6\% per annum compounded monthly?

1. ₹10,00,500
2. ₹10,50,000
3. ₹10,00,000
4. ₹10,05,000

Options 1. 1
2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 Match List-I with List-II (Given that c is an arbitrary constant)

List-I

List-II

(A) $\int \frac{dx}{\sqrt{a^2 - x^2}}$ (I) $\log_e |x + \sqrt{x^2 - a^2}| + c$

(B) $\int \sqrt{a^2 - x^2} dx$ (II) $\sin^{-1} \frac{x}{a} + c$

(C) $\int \sqrt{x^2 - a^2} dx$ (III) $\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + c$

(D) $\int \frac{dx}{\sqrt{x^2 - a^2}}$ (IV) $\frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log_e |x + \sqrt{x^2 - a^2}| + c$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (II), (C) – (IV), (D) – (I)
2. (A) – (IV), (B) – (II), (C) – (I), (D) – (III)
3. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
4. (A) – (II), (B) – (III), (C) – (IV), (D) – (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.52 If $\vec{a}$, $\vec{b}$ and $\sqrt{3}\vec{a} - \vec{b}$ are three unit vectors, then the angle between $\vec{a}$ and $\vec{b}$ is:

1. $\frac{\pi}{6}$
2. $\frac{\pi}{4}$
3. $\frac{\pi}{3}$
4. $\frac{\pi}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.53 If $f(x) = \begin{cases} \frac{\tan(\frac{\pi}{4} - x)}{\cot 2x}, & x \neq \frac{\pi}{4}, \\ 2K + 1, & x = \frac{\pi}{4}, \end{cases}$ is continuous at $x = \frac{\pi}{4}$, then the value of K is equal to

1. $\frac{1}{4}$
2. $\frac{1}{2}$
3. $-\frac{1}{2}$
4. $-\frac{1}{4}$

Options 1. 1

2. 2
3. 3
4. 4

Q.54 A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and are found to be heart. Then the probability of the missing card to be a heart is:

1. $\frac{14}{25}$
2. $\frac{11}{25}$
3. $\frac{39}{50}$
4. $\frac{11}{50}$

Options 1. 1

2. 2
3. 3
4. 4

Q.55 Let E and F be two events such that $P(E) = \frac{1}{3}$, $P(F) = \frac{1}{4}$ and $P(E \cap F) = \frac{1}{5}$.

Then the value of $P(F | E)$ is equal to

1. $\frac{4}{5}$
2. 1
3. $\frac{3}{5}$
4. $\frac{1}{12}$

Options 1. 1

2. 2

3. 3

4. 4

Q.56

The acute angle between the lines $\vec{r} = (4\hat{i} - \hat{j}) + \lambda(2\hat{i} + \hat{j} - 3\hat{k})$ and $\frac{x-1}{1} = \frac{y+1}{-3} = \frac{z-2}{2}$ is

1. $\frac{\pi}{6}$
2. $\frac{\pi}{4}$
3. $\frac{\pi}{3}$
4. $\frac{\pi}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.57

If $\begin{bmatrix} x+y & 4 \\ 1+z & y \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$, then

1. $x = 4, y = -6, z = 4$
2. $x = -4, y = 6, z = 4$
3. $x = -4, y = 6, z = -4$
4. $x = 4, y = -6, z = -4$

Options 1. 1

2. 2

3. 3

4. 4

Q.58

The value of $\int \left(\frac{1}{\log_e x} - \frac{1}{(\log_e x)^2} \right) dx$ is

1. $x \log_e x + c$: where c is an arbitrary constant
2. $e^{x \log_e x} + c$: where c is an arbitrary constant
3. $\frac{e^x}{\log_e x} + c$: where c is an arbitrary constant
4. $\frac{x}{\log_e x} + c$: where c is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.59

Match List-I with List-II

List-I**List-II**

(A) Maximum value of $f(x) = \sin^2 x - \cos^2 x \forall x \in (\pi, 2\pi)$ (I) 0

(B) Minimum value of $f(x) = \sin x \cdot \cos x$

(II) 1

(C) Point of Minima of $f(x) = x^x (x > 0)$

(III) $-\frac{1}{2}$

(D) Maximum value of $f(x) = -x^{2026}$

(IV) $\frac{1}{e}$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.60 The area (in square units) bounded by the curve $y = \cos x$ between $x = 0$ and $x = 2\pi$ in first quadrant is equal to:

1. 4
2. 2
3. 1
4. 3

Options 1. 1

2. 2
3. 3
4. 4

Q.61 The interval in which the function $f(x) = 2x^3 + 3x^2 - 12x + 1$ is strictly increasing, is

1. $(-\infty, -2) \cup (1, \infty)$
2. $(-\infty, 1)$
3. $(-\infty, -1) \cup (2, \infty)$
4. $(-2, 1)$

Options 1. 1

2. 2
3. 3
4. 4

Q.62 A integrating factor of the differential equation $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2} (x > 0)$ is equal to

1. x
2. $-x$
3. $\frac{1}{x}$
4. $-\frac{1}{x}$

Options 1. 1

2. 2
3. 3
4. 4

Q.63 The area (in square units) of the region bounded by the curve $x^2 = y$ and the straight line $y = 4$ in the first quadrant is equal to

1. $\frac{3}{2}$
2. $\frac{16}{3}$
3. $\frac{32}{3}$
4. $\frac{7}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.64 The value of $\int_0^{\pi/2} \log_e \left(\frac{5 + 2 \sin x}{5 + 2 \cos x} \right) dx$ is

1. 0
2. 1
3. 2
4. π

Options 1. 1

2. 2
3. 3
4. 4

Q.65 Let N be set of natural numbers, then the function $f : N \rightarrow N$, defined by

$$f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$$

1. injective but not surjective
2. surjective but not injective
3. bijective
4. neither injective nor surjective

Options 1. 1

2. 2

3. 3

4. 4

Q.66 Let $\mathbf{A}$ be any square matrix of order n , then which of the following are true?

(A) $|\text{adj}\mathbf{A}| = |\mathbf{A}|^{n-1}$

(B) $|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|}$

(C) $|\text{adj}\mathbf{A}| = |\mathbf{A}|^n$

(D) $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (D) only
3. (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.67 The point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate, is

1. (2, -4)
2. (2, 4)
3. $(1, 2\sqrt{2})$
4. $(1, -2\sqrt{2})$

Options 1. 1

2. 2

3. 3

4. 4

Q.68 The greatest integer function $f(x) = [x]$ is differentiable for all values of

1. $x \in \mathbf{Z}$ (set of integers)
2. $x \in \mathbf{R}$ (set of real number)
3. $x \in \mathbf{R} - \mathbf{Z}$
4. $x \in \mathbf{Q}$ (set of rational number)

Options 1. 1

2. 2

3. 3

4. 4

Q.69 Let the matrix $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$. Then which of the following are true?

(A) $\text{adj}\mathbf{A} = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$

(B) $\det(\mathbf{A}) = 5$

(C) $\det(\text{adj}A) = 25$

(D) If $A^3 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $a + b = c + d$

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A), (B) and (C) only
3. (B) and (D) only
4. (A), (B) and (D) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.70 Let $|\vec{a}| = 5$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 6$, then the value of $|\vec{a} \times \vec{b}|$ is equal to

1. 8
2. 6
3. 10
4. 60

- Options 1. 1
2. 2
3. 3
4. 4

Q.71 If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, where $|x| < 1$, $|y| < 1$ and $x \neq y$, then

1. $\frac{dy}{dx} = \sqrt{\frac{1+y}{1+x}}$
2. $(1+x)^2 \frac{dy}{dx} + 1 = 0$
3. $(1+x^2) \frac{dy}{dx} + 1 = 0$
4. $(1+x)^2 \frac{dy}{dx} = 1$

- Options 1. 1
2. 2
3. 3
4. 4

Q.72 The general solution of the differential equation $x(1+y^2)dx + y(1+x^2)dy = 0$ is

1. $(1+y^2)(1+x^2) = C$: where C is an arbitrary constant
2. $(1-y^2)(1-x^2) = C$: where C is an arbitrary constant
3. $(1-y^2)(1+x^2) = C$: where C is an arbitrary constant
4. $(1+y^2)(1-x^2) = C$: where C is an arbitrary constant

- Options 1. 1
2. 2
3. 3
4. 4

Q.73 A can solve 90% problems and B can solve 70% problems of the book. A problem is selected at random from the book. The probability that the problem is solved, is equal to

1. 0.87
2. 0.85
3. 0.63
4. 0.97

- Options 1. 1
2. 2
3. 3
4. 4

Q.74 If A and B are two independent events, then which of the following is/are true?

(A) $P(A \cap B) = 0$

(B) $P(A \cup B) = 1 - P(A')P(B')$

(C) $P(A \cup B) = P(A)P(B)$

(D) $P(A \cap B) = P(A)P(B)$

Choose the correct answer from the options given below:

1. (B) and (D) only
2. (A) and (C) only
3. (B) only
4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.75 Match List-I with List-II

List-I

(A) The vectors $\lambda\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + \lambda\hat{j} + \hat{k}$ are perpendicular if λ is equal to

(B) The vectors $3\hat{i} + 6\hat{j} - \hat{k}$ and $2\hat{i} + \lambda\hat{j} - 2\hat{k}$ are collinear if λ is equal to

(C) The number of vectors of unit-length which are perpendicular to both the vectors $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 5\hat{k}$ is

(D) If $|\vec{a}| = 1$ and $\vec{a} + \vec{b} = \vec{0}$, then $|\vec{b}|$ is equal to

List-II

(I) 1

(II) -1

(III) $2/3$

(IV) 2

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

2. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)

3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.76 If R be a relation on the set of integers $\mathbb{Z}$, given by $R = \{(a, b) : (a - b) \text{ is a multiple of } 3\}$, then R is:

1. Reflexive, Symmetric but not Transitive
2. Reflexive, Transitive but not Symmetric
3. Symmetric, Transitive, but not Reflexive
4. an equivalence relation

Options 1. 1

2. 2

3. 3

4. 4

Q.77 For some constant 'k', if the system of linear equations

$$2x - y + 3z = 1$$

$$x - 2y + z = 3$$

$$kx + y - z = 0$$

has a unique solution, then

1. $k = -\frac{4}{5}$
2. $k \neq -2$
3. $k \neq 2$
4. $k \neq -\frac{4}{5}$

Options 1. 1

2. 2
3. 3
4. 4

Q.78 Match List-I with List-II

List-I

List-II

(Inverse Trigonometric Function) (Principal Value)

(A) $\sin^{-1}\left(-\frac{1}{2}\right)$

(I) $\pi/6$

(B) $\cos^{-1}\left(-\frac{1}{2}\right)$

(II) $-\pi/6$

(C) $\tan^{-1}(-\sqrt{3})$

(III) $2\pi/3$

(D) $\cot^{-1}(\sqrt{3})$

(IV) $-\pi/3$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.79

If $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$, where a, b and c are non zero constants, then the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is

1. -1
2. 1
3. 0
4. abc

Options 1. 1

2. 2
3. 3
4. 4

Q.80

The value of k for which the lines $\frac{2x-3}{4} = \frac{3-y}{k} = \frac{z-2}{-2}$ and $\frac{x-2}{1} = \frac{y}{4} = \frac{5-z}{3}$ are perpendicular to each other is:

1. $\frac{5}{2}$
2. $-\frac{5}{2}$
3. 2
4. -2

Options 1. 1

2. 2
3. 3
4. 4

Q.81

The corner points of the bounded feasible region determined by the system of linear constraints are $(0, 0), (5, 0), (3, 4)$

and $(0, 5)$. Let $Z = px + qy$ where $p, q > 0$. Condition on p and q so that the maximum of Z occurs at both $(5, 0)$ and $(3, 4)$

is

1. $3q = p$
2. $q = p$
3. $2q = p$
4. $2p = q$

Options 1. 1

2. 2
3. 3
4. 4

Q.82

If $\vec{a}, \vec{b}, \vec{c}$ are vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 1, |\vec{b}| = 2, |\vec{c}| = 5$, then the expression $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ equals

1. 30
2. -15
3. 0
4. 1

Options 1. 1

2. 2
3. 3
4. 4

Q.83

If the matrix $A = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$, then the value of $\det(A^2 - 2A)$ is equal to

1. 5
2. -5
3. 25
4. -25

Options 1. 1

2. 2
3. 3
4. 4

Q.84

Consider a L.P.P, Maximize $Z = 3x + 5y$ subject to constraints $2x + 6y \leq 6, x - y \geq 0, x \geq 0, y \geq 0$. Then which of the following are true?

- (A) The feasible region of L.P.P is bounded region
- (B) The corner points of the feasible region are $(0, 0), (2, 2), (0, 1)$
- (C) Maximum value of Z is 9
- (D) Point $(1, 3)$ lies in the feasible region

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) and (C) only
3. (A) and (D) only
4. (A) and (B) only

Options 1. 1

2. 2
3. 3
4. 4

Q.85 If a line makes angles α, β and γ with the positive directions of coordinate axes respectively, then

$\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is equal to

1. 1
2. -1
3. 2
4. -2

Options

1. 1
2. 2
3. 3
4. 4